IRON MAN GYM ADMIN

Project report and feature guide

A gym management system with staff tools and a member-facing Progressive Web App (PWA). This report describes the prototype, its workflows, technical structure, and deployment considerations.

Project profile	Details
Application	Gym administration and member self-service
Technology	Express 5, EJS, SQLite via node:sqlite, and CSS
Languages	Arabic and English interface; English project documentation
Release state	Prototype with demo records and simulated integrations
Report date	October 25, 2026

Privacy note: This public edition contains no real staff or member names. The original Arabic report and real local records should remain on the owner's device.

Contents

- Product overview and user roles
- Attendance, memberships, payments, and the member app
- Training, health tracking, notifications, and inventory
- Architecture, setup, privacy, and production readiness

1. Product overview

The application brings day-to-day gym operations into one web system. Staff manage members, subscriptions, attendance, payments, training plans, products, cleaning tasks, and announcements. Members use a mobile-friendly PWA to review their own account, schedule, progress, and notifications.

Access and roles

Role	Main access
Administrator	Full dashboard, staff, schedules, plans, products, imports, member records, and financial totals.
Reception	Check-in, member records, subscriptions, payments, and shop sales. Financial dashboard totals are restricted.
Trainer	Assigned members, workout programs, templates, and InBody records.
Cleaner	Daily cleaning checklist and work-time clock.
Member	Their own account, attendance, workouts, progress, notifications, and product catalog.

Authorization is checked on the server. Hiding a button in the interface is not the security boundary; route permissions are enforced in the middleware.

2. Attendance and membership operations

The gym day begins at 6:00 AM. Activity between midnight and 6:00 AM is assigned to the previous gym day, which keeps overnight visits together with the evening session that started them.

Day	Iron gym	Kickboxing
Saturday-Wednesday	3:30 PM-6:00 AM	7:30 PM-9:00 PM on Sunday and Tuesday
Thursday	3:30 PM-1:00 AM	7:30 PM-9:00 PM
Friday	6:00 PM-9:00 PM	Closed

Hours are configurable in the schedule page. Before accepting an attendance event, the application checks the member's active subscription and the applicable opening or class window.

Fingerprint workflow

- The first fingerprint event starts a visit; the next event closes it and records the session duration.
- Iron gym and kickboxing access are checked separately. Iron members may make multiple visits per day; kickboxing is limited to one visit per day.
- A member needs a fingerprint ID on their record. The prototype uses a simulation button until a real device integration is configured.
- If a member forgets to check out, the active timer stops after five hours and the incomplete visit is not used as a completed session.

Member records and payments

Member profiles can include contact and identity fields, subscription status, trainer assignment, fingerprint ID, notes, goals, and workout details. National IDs are limited to administrator and reception roles; trainers do not see them.

- Members can have separate iron gym and kickboxing subscriptions.
- Early renewal starts after the current subscription ends. Changing a plan's price does not rewrite existing subscriptions.

- The balance is calculated from subscriptions and account purchases minus recorded payments. Partial payments are supported.
- The member profile brings subscription dates, balances, attendance, purchases, workout records, and InBody history together.

3. Member app and training tools

The member interface is a PWA that can be added from a mobile browser without an app-store installation. Members can activate an account using the gym's existing registration details and a one-time verification code.

Member app areas

Area	What it provides
Home	Subscription days remaining, current visit timer, upcoming classes, gym notices, and recent notifications.
My program	The member's assigned workout plan, with a private copy that can be updated by the member and trainer.
Progress	Attendance and workout adherence, body-composition measurements, weight history, and progress charts.
Account	Profile, balance, purchases, visit history, notification preferences, and password changes.
Shop	Product catalog and current prices.

Workout plans

- Eight ready-made templates are included, covering full-body plans, PPL splits, two-muscle sessions, and an Arnold split.
- Templates can be created from scratch or copied from an existing plan. Assigning a plan creates a member-specific copy.
- The trainer and member can edit the assigned copy. Updates do not alter the source template, and the other party receives a notification.
- The member records sets and weights; the app shows prior performance and personal records.

Health and adherence

Staff can enter InBody readings, including weight, skeletal muscle mass, body-fat mass and percentage, BMI, visceral fat, basal metabolic rate, total body water, and score. The application compares readings over time and displays charts.

Trainers can set a goal, target date, weekly training days, calories, protein, carbohydrates, fat, and water. Members can record daily nutrition adherence, water intake, and optional weight.

Weekly adherence component	Weight
Attendance compared with required training days	40%
Days with workout sets recorded	30%
Nutrition adherence (yes = 1, partial = 0.5)	30%

4. Notifications, shop, and staff tools

Notifications are checked by a background task every ten minutes. Each notification has a unique key to avoid sending the same reminder twice. SMS is an integration point and is printed to the console in demo mode.

Recipient	Examples and timing
Member	Subscription reminders before and after expiry; schedule changes; class reminders; inactivity reminders after seven days; birthdays, payments, purchases, plan changes, and InBody updates.
Trainer	New assigned members, a member's absence, and program changes.
Administrator and reception	Daily summary at 10:00 AM.
Administrator	Low stock, supplements approaching expiry, and completed cleaning tasks.

Members can turn off selected class, inactivity, and promotional reminders. Subscription and gym-closure notices remain enabled.

Shop and inventory

- Products are grouped into supplements, accessories, and drinks, with sale price, purchase price, quantity, and optional expiry date.
- A sale can be paid in cash or charged to the member's account. The system prevents sales above available stock.
- Cancelling a sale restores the sold quantity. Administrators can review monthly sales and profit, and receive low-stock or expiry alerts.

Cleaning and staff

The cleaner role sees a daily task list and work-time clock. Administrators can add or remove tasks. Staff with a fingerprint ID can record shift start and end; monthly work hours appear in the staff page.

Member import

Staff can import member rows from CSV. The import page includes a sample format, rejects duplicate records, and reports row-level problems. Replace all sample values before using the template for real data.

5. Technology and project layout

Component	Responsibility
app.js	Express setup, middleware, localization, and route registration.
db.js	SQLite schema, indexes, and initial defaults.
core.js, dates.js, schedule.js	Business rules, gym-day calculations, attendance, and opening hours.
routes/ and views/	Authentication, staff, member, program, shop routes, and EJS pages.
notify.js	Notification scheduling and SMS provider integration point.
seed.js	Synthetic Arabic demo data for local exploration.
locales/en.json	English interface translations; Arabic is the default interface language.
test/	Automated application, date, and schedule tests.

6. Setup, privacy, and production readiness

Requirements

Use Node.js 22.13 or later. Run `npm install`, then `npm run seed` to create synthetic demo records, and `npm start` to launch the app at `http://localhost:3000`. Run `npm test` to execute the automated tests. Use `npm run seed -- --force` only when you intend to replace the current database with demo data.

Prototype limitations

- SMS delivery is not connected; messages appear in the server console.
- OTP codes are shown on screen in demo mode. Set `DEMO=0` to hide them and the simulated member fingerprint button.
- Fingerprint hardware is not integrated. Confirm the device model and vendor interface before connecting real attendance data.
- Database migrations are not included. Add migrations and tested backups before importing real records.

Privacy rules for this public repository

- The published seed data uses generic Arabic demo labels rather than names of real people.
- The original Arabic report and the local SQLite database are excluded from the public release. The English report contains no real staff or member names.
- Keep real member information, credentials, fingerprint IDs, health readings, and private reports on a protected local system. Never commit them to Git.
- Replace demo passwords and add a strong `SESSION_SECRET` before any real deployment.

Production checklist

- Connect an SMS provider and test delivery, retry behavior, and consent requirements.
- Integrate and test the fingerprint device without exposing biometric data in logs or source control.
- Deploy behind HTTPS with persistent storage for SQLite, access controls, backup/restore procedures, monitoring, and database migrations.
- Remove demo accounts and sample records before inviting real staff or members.

This document is an English-language summary of the project. Product interface labels remain available in Arabic and English.